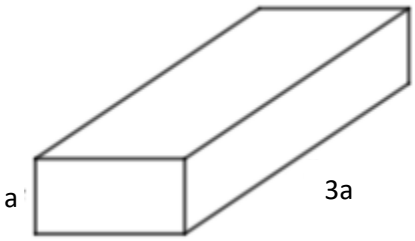
## GRADE 10 REVISION: MEASUREMENT

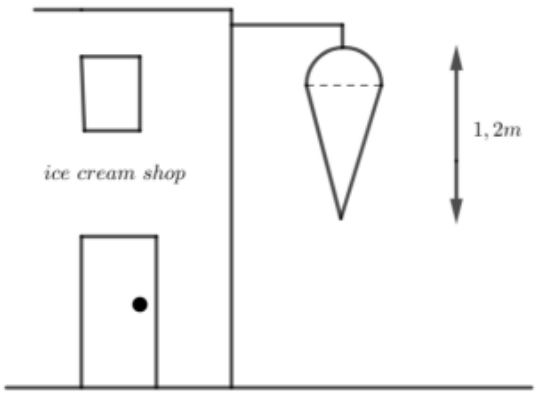
13.2 A rectangular prism has sides $3a$, $2a$, and $a$ centimetres. The total surface area of the prism is $550cm^2$.

13.2.1 Determine the value of $a$. (4)

13.2.2 If all sides are multiplied by a factor of $k$ the total surface area reduces to $137{,}5cm^2$. Determine the value of $k$. (4)




13.4 An ice cream shop displays a large metal ice cream advertising its business. The shape is hollow and made up of a cone and a hemisphere as shown. The vertical height of the display from top to bottom is 1,2 metres and the radius of the hemisphere is 0,3 metres.




13.4.1 Show that the total volume of the metal ice cream is $\frac{9}{200}\pi$ cubic metres. (4)

13.4.2 Calculate the total surface area of the metal ice cream. (4)

Useful formulae: $V_S = \frac{4}{3}\pi r^3$ , $SA_S = 4\pi r^2$ , $V_C = \frac{1}{3}\pi r^2 h$ , $SA_C(open) = \pi rs$